User Manual

UIM344 / UIM344S / UIM344XS Controller
with Pulse-Direction Control Interface
for Closed-Loop Integrated Stepper Motor

UIM20xxPM / UIM2040PM

UIM28xxPM / UIM2852PM

UIM42xxPM / UIM4247PM

UIM57xxPM / UIM5756PM / UIM5776PM

UIM57xxPMB / UIM5756PMB / UIM5776PMB

UIM86xxPM / UIM8696PM

UIM86xxPMB / UIM8696PMB

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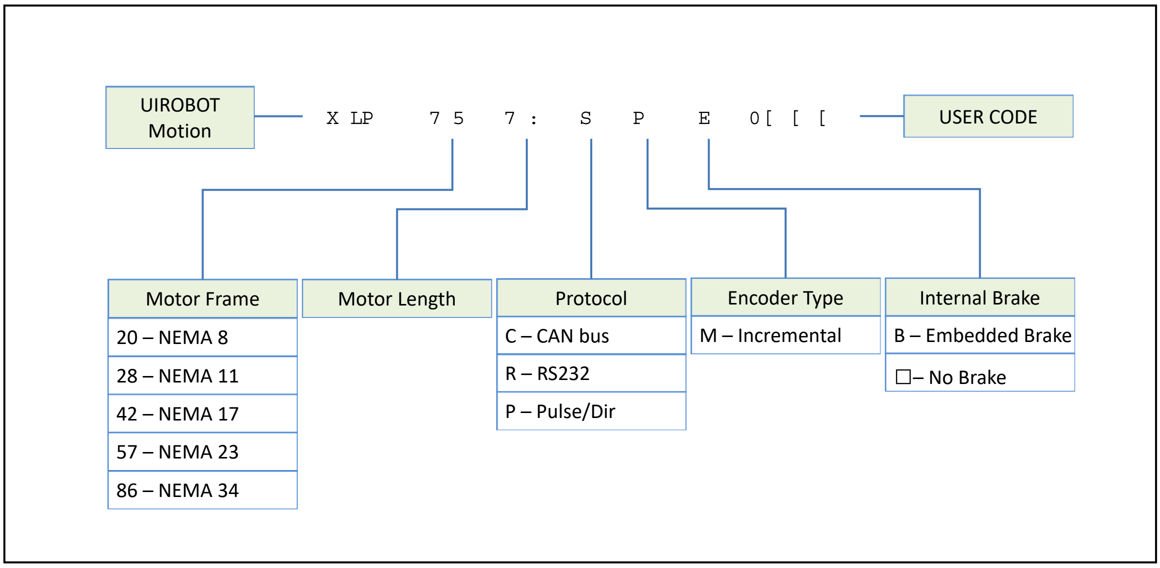
[Contact information]

United Intelligence Robotics (Zhejiang)Co.,Ltd.
Add: Bldg B3, Innovation Park, Pinghu, Jiaxing City, Zhejiang Province, China 314200
Tel: (086)0573-85235951
Email: support@uirobot-zj.com

Web: www.uirobot.com

UIM344 Serial Integrated Stepper Motor Order Code



Revision

Manual version	Revision date	Change
V 1.0	November 1st, 2023	Initial version

UIM344/344S/344XS
Closed-loop Integrated Stepper Motors with Pulse-Direction Control Interface

UIM344 series Integrated Stepper Motors are comprised of stepper motor, encoder, driver, and closed-loop controller. These motors interfaces with user controller / PLC / computer through 3 wired signals: Pulse, direction and enable. Motor frame sizes offered are NEMA 8 / 11 / 17 / 23 / 34 / 42. Depending on motor size, rated current is from 1 Amp to 8 Amp, and supply voltage is from 12V to 48V.

With the embedded encoder and closed-loop controller, these motors can perform an accurate positioning motor even under the stalling situation. Missing steps compensation are done by the motor itself.

The fully enclosed and die-cast aluminum body provides a rugged, durable protection and improves heat dissipation.

Closed-Loop Controller

- Closed-loop Control
- Missing step self-compensation
- High positioning accuracy, high response, max speed 3000 rpm (depends on specific motor type)
- Aluminum alloy casing, sturdy and durable for better heat dissipation
- Size as small as 20 mm x 20 mm x 16 mm

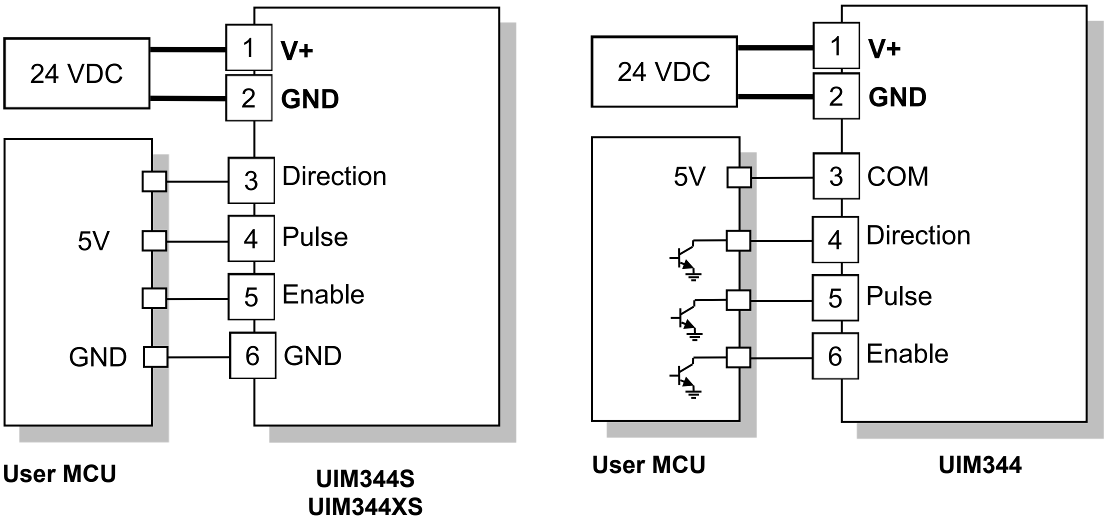
Motor Driver

- Wide voltage input 12 ~ 48 VDC (depends on specific motor type)
- Up to 8 A phase current, adjustable
- PWM constant current
- Micro-stepping, 1 ~ 1/128 resolution
- No vibration/noise at low speed
- Overcurrent/overvoltage/overheating protection

Contents

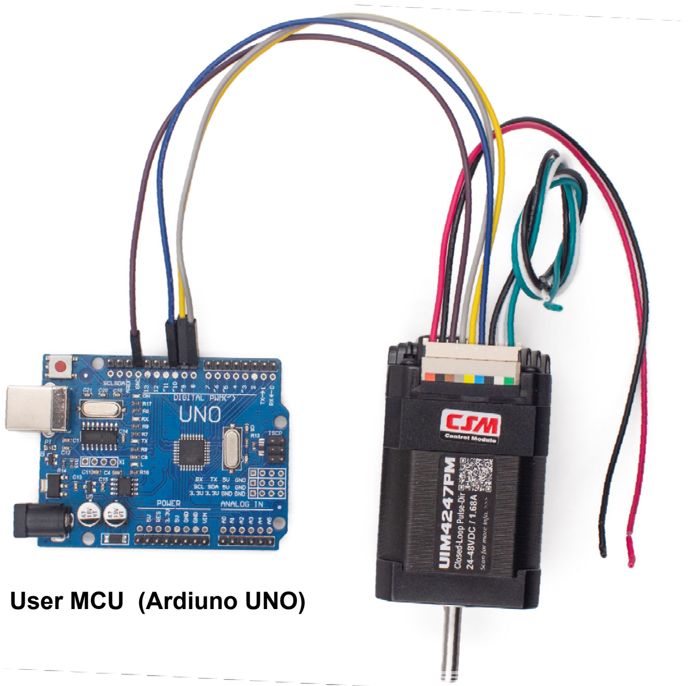
Typical Application

Figure 0-1: Typical Application for UIM344



Note:

As shown below, the Pulses, Direction and Enable signals should be provided by the User MCU to control the motor driver ON / OFF, speed, acceleration and deceleration and position.



User MCU (Ardiuno UNO)

Motor: UIM4247PM

Visit www.uirobot.com to get the free sample code for programing above Arduino to control the motor.

Control Terminal Description

No.	Name	Description
1	V+	Supply voltage, 24 – 48 VDC (for UIM344), 12 – 24 VDC (for UIM344S/UIM344XS).
2	GND	Supply voltage ground.
3	COM	Opto-coupler common port (Not available for UIM344S / UIM344XS)
4	DIR	Direction Input, Opto-couple active (for UIM344) or High Level (for UIM344S/344XS) to run in CW; otherwise, run in CCW.
5	STP	Stepping pulse Input, pulse duration should > 4μs
6	ENA	Enable the Controller, Opto-couple active (for UIM344) or High Level (for UIM344XS/344S) to enable the driver; otherwise, the driver will stay disabled.
7	GND	Internally linked to Terminal 2.
8	TX	Transmit (UIM344 to PC) Terminal of the UART interface used to config the Micro-stepping / Working Current and Idel Current, etc.
9	RX	Receive (PC to UIM344) Terminal of the UART interface used to config the Micro-stepping / Working Current and Idel Current, etc.

Pulse Rate vs Speed

Following is the equation to calculate the required pulse rate (PPS, pulse per second):

$$RPM = 60 \times \frac{PPS}{MCS \text{ x } 200}$$

or

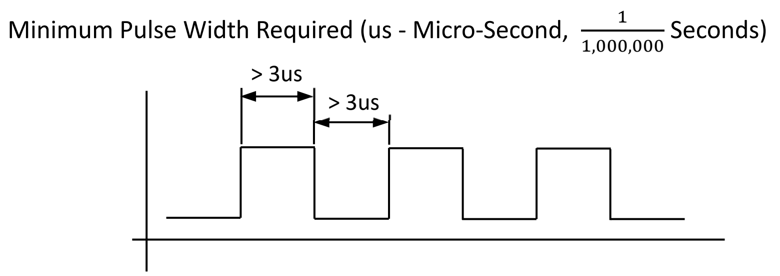
$$PPS = \frac{RPM \times MCS \times 200}{60}$$

Where:

MCS - Micro-stepping Resolution, e.g., 1, 2, 4, 8, 16, 32, 64, etc.;
RPM - Revolutions Per Minute, motor speed;
PPS - Pulses Per Second, sent to the motor:

Pulse requirements

Minimum pulse width is required to be larger than 3 micro-seconds for both high and low level periods.



Motor Connection Guide

Figure 0-2: UIM344XS Wiring Diagram (Example Motor Model: UIM2040PM)

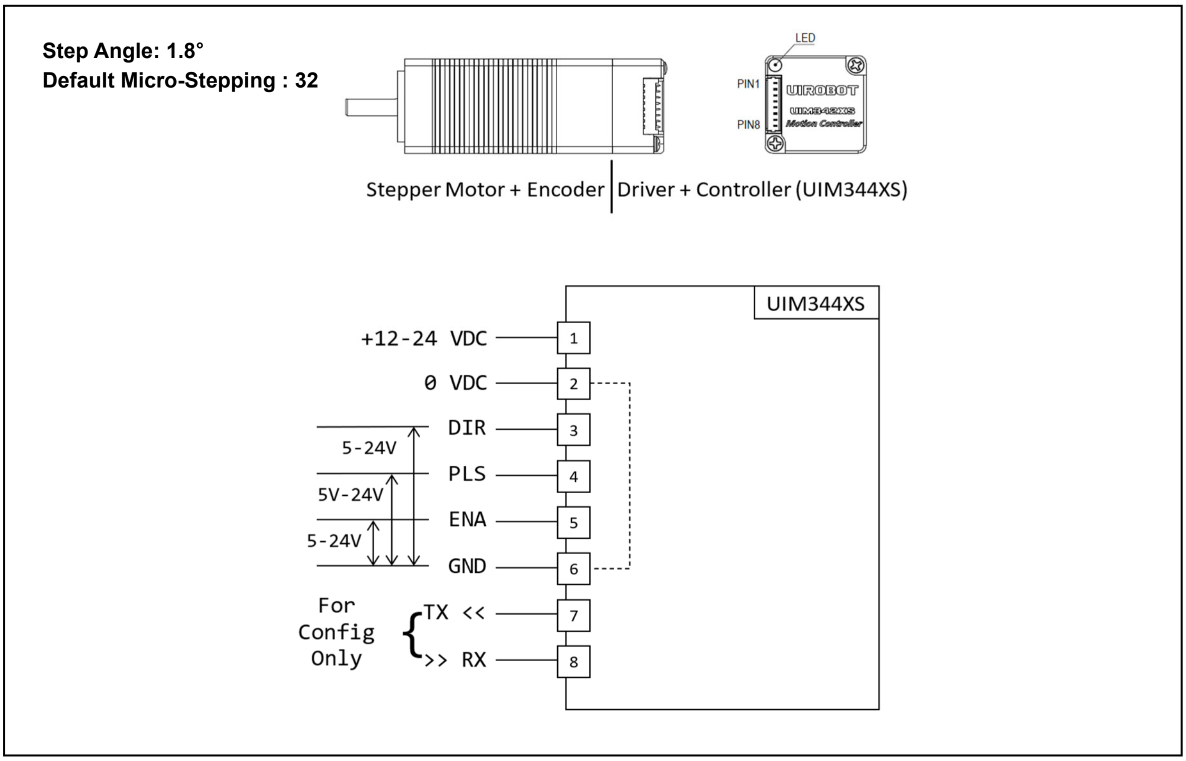


Figure 0-3: UIM344S Wiring Diagram (Example Motor Model: UIM2852PM)

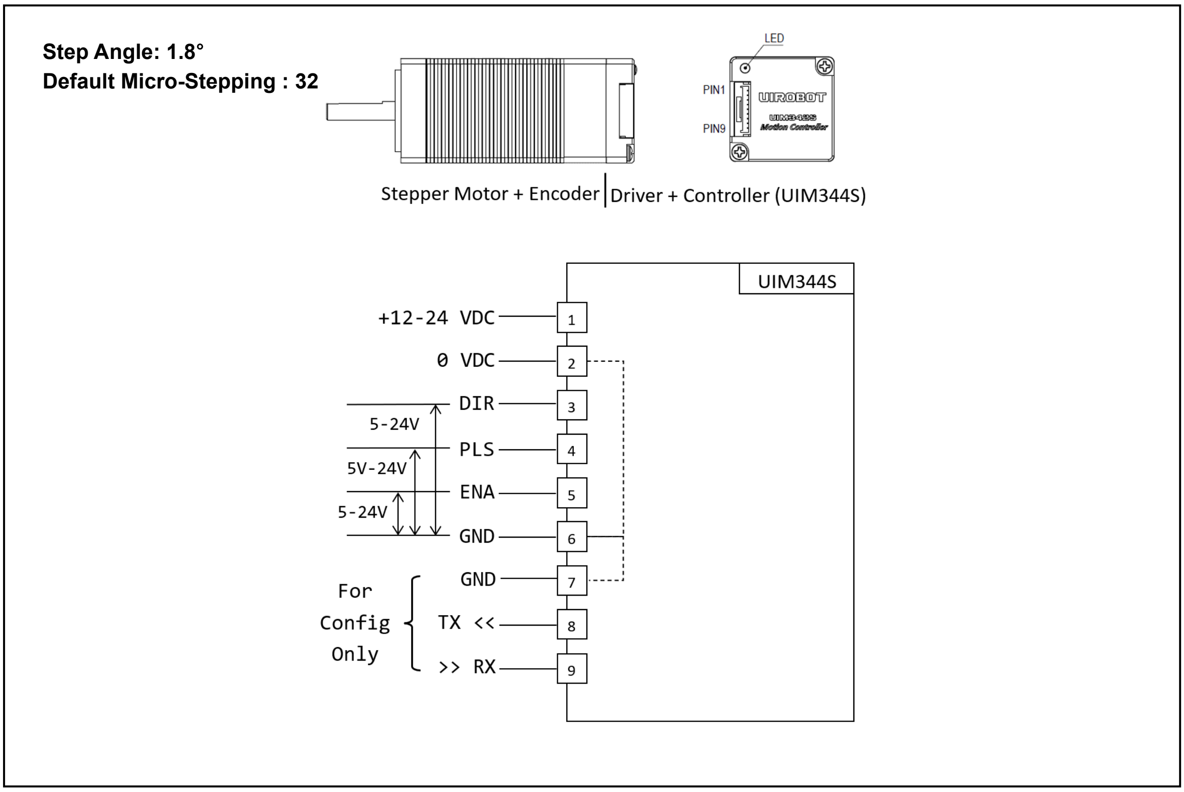


Figure 0-4: UIM344 Wiring Diagram (Example Motor Model: UIM4247PM)

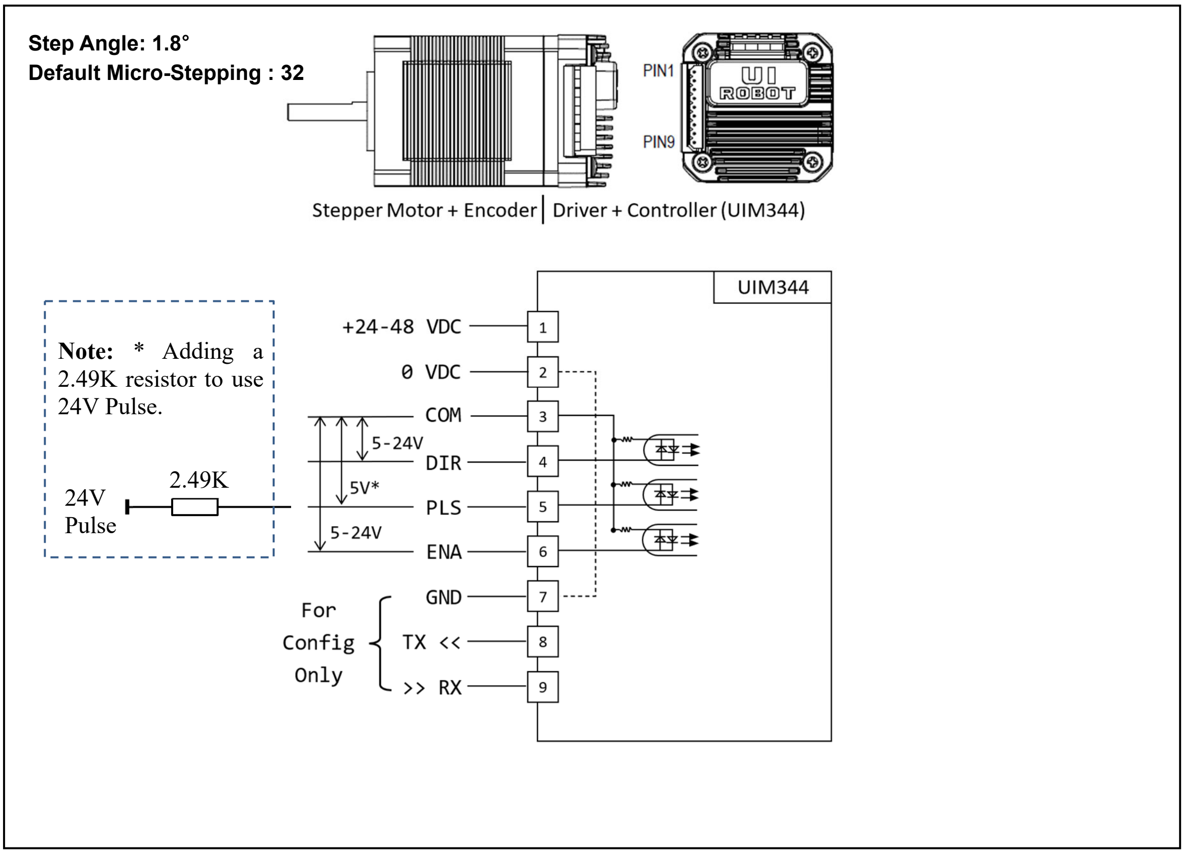


Figure 0-5: UIM344 Wiring Diagram with Brake (Example Motor Model: UIM4247PMB)

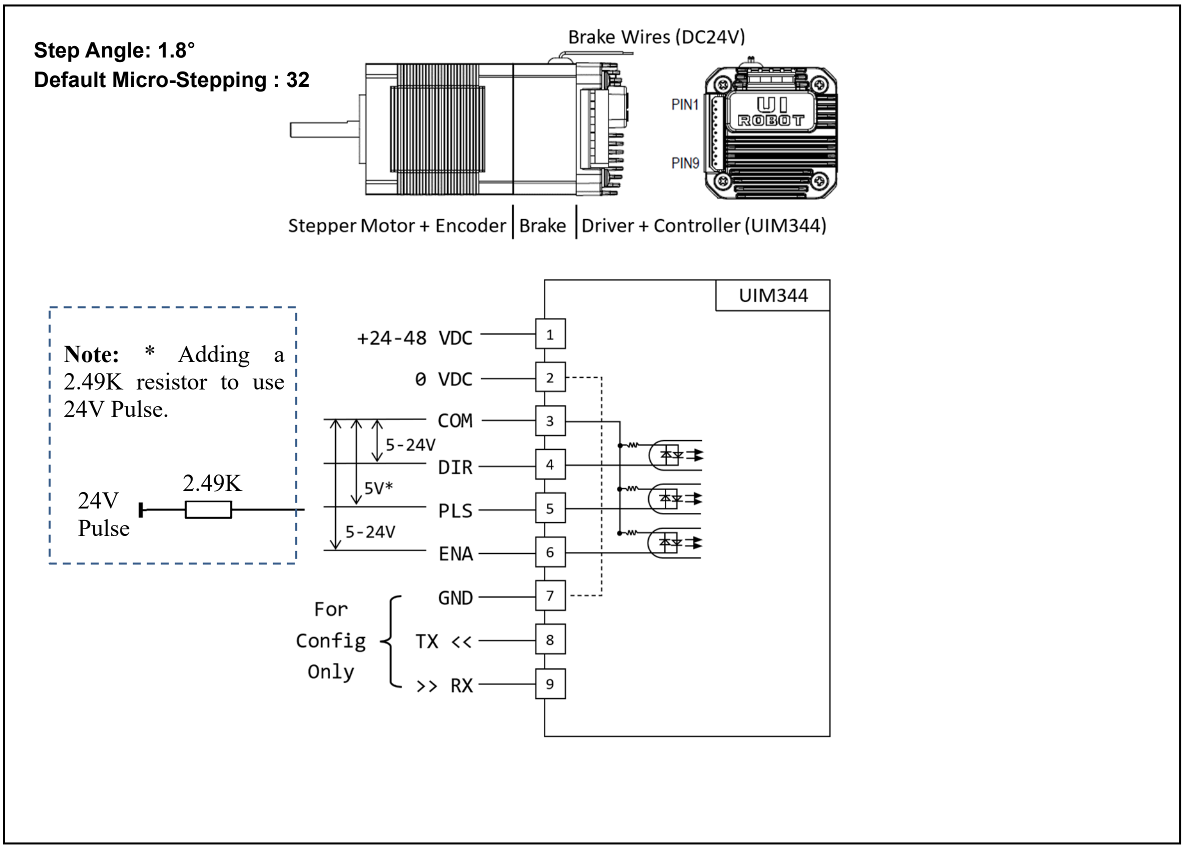


Figure 0-6: UIM344 Wiring Diagram (Example Motor Model: UIM5756PM)

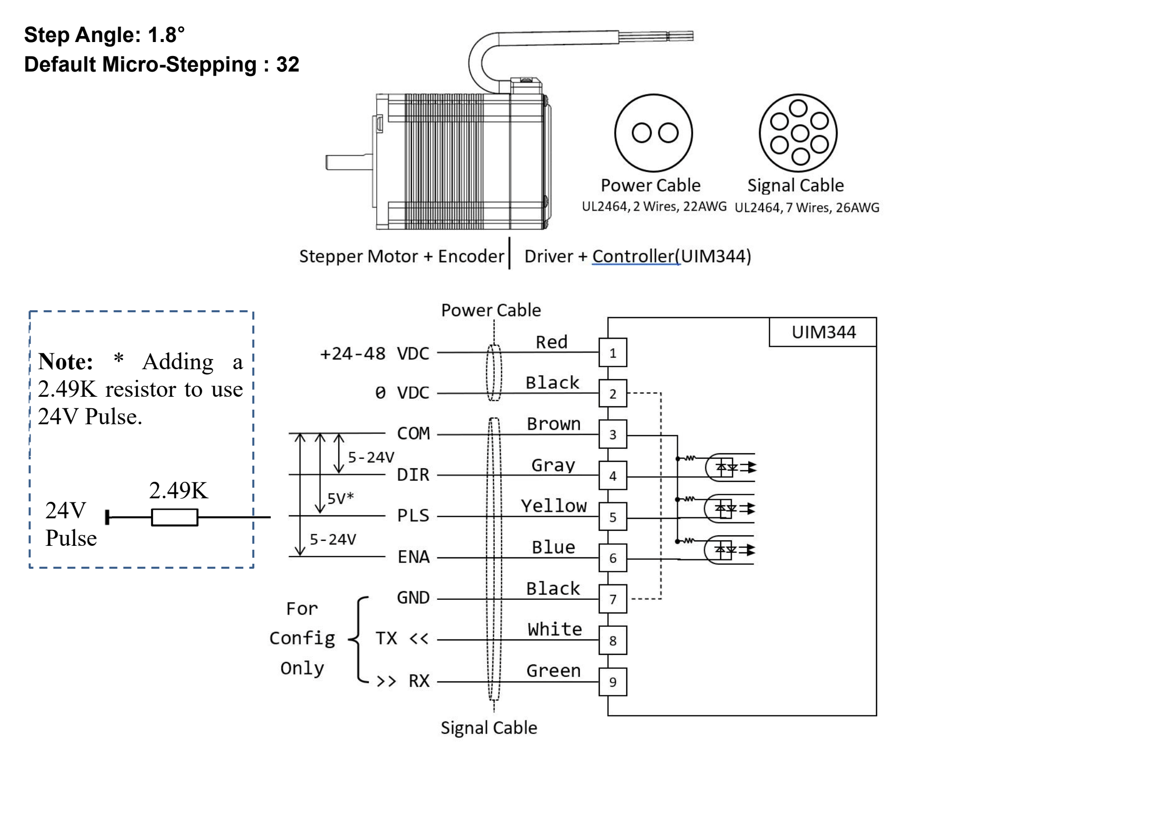


Figure 0-7: UIM344 Wiring Diagram with Brake (Motor Model: UIM5756PMB)

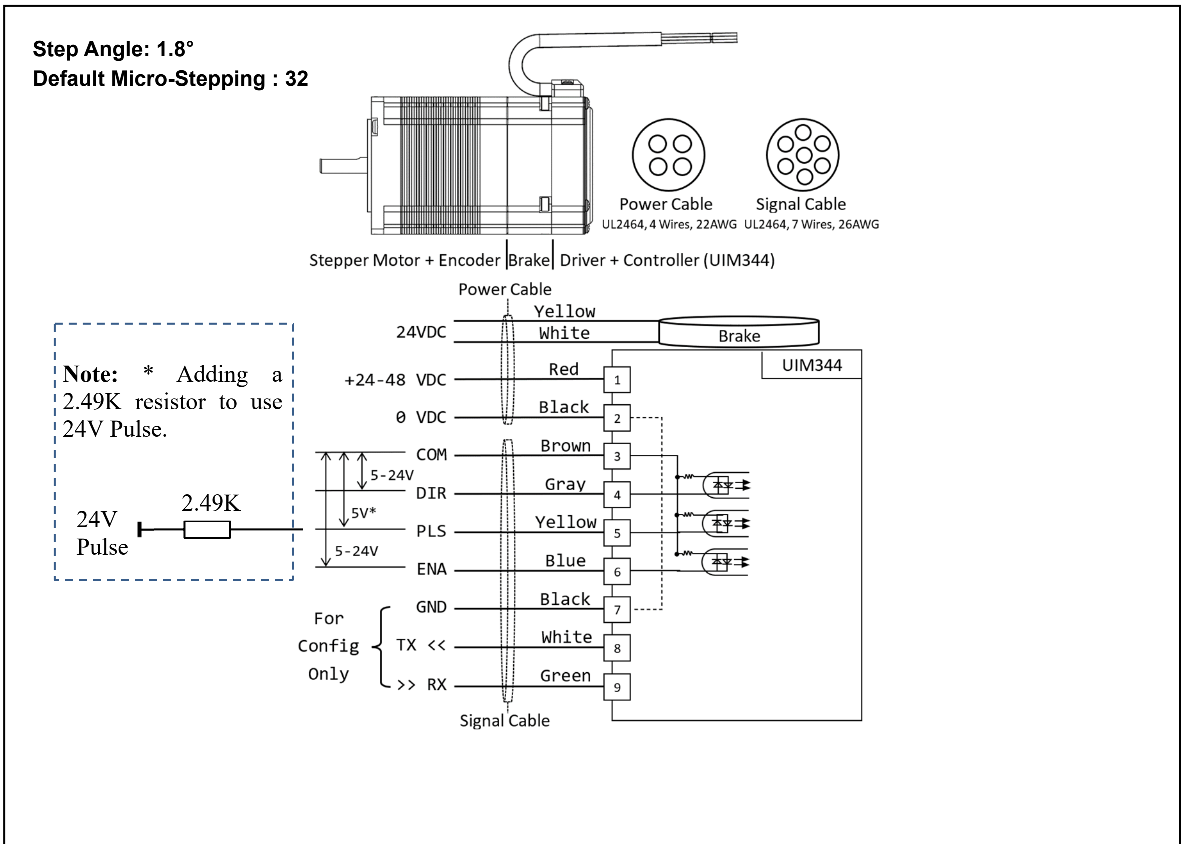


Figure 0-8: UIM344 Wiring Diagram (Example Motor Model: UIM8696PM)

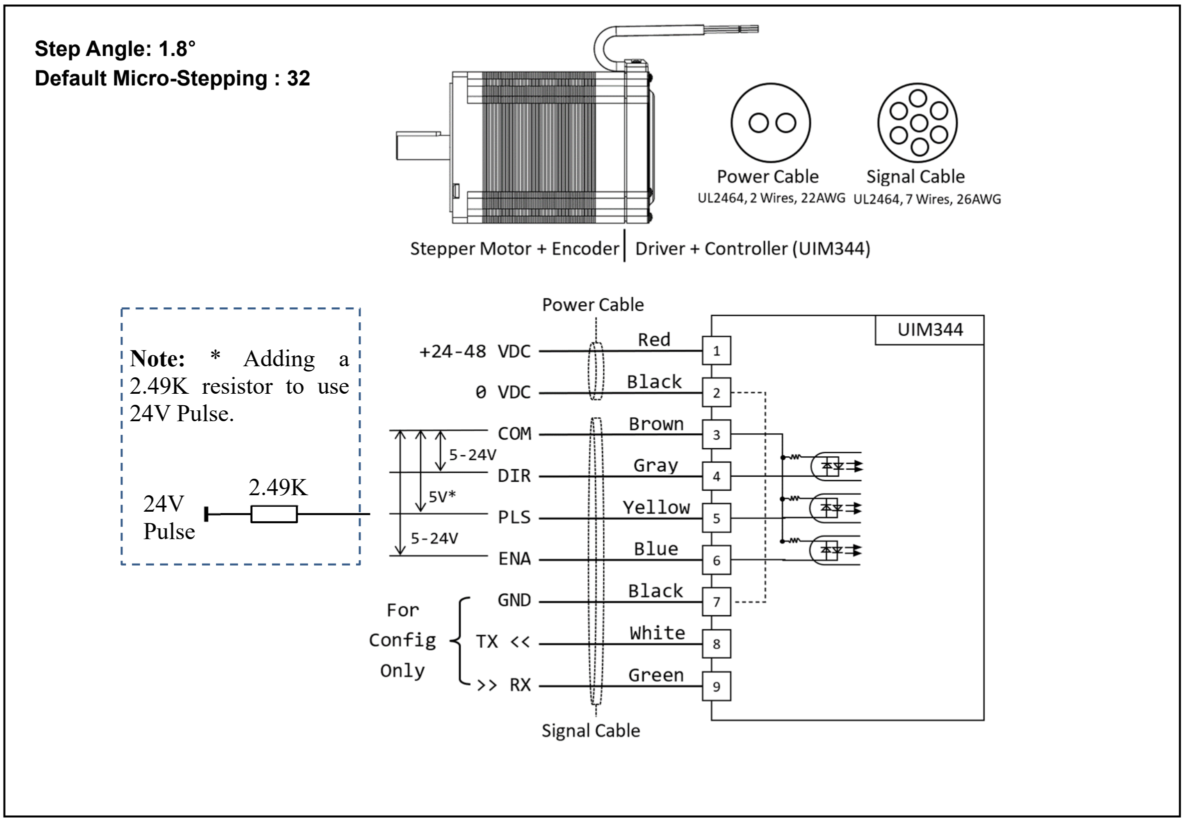
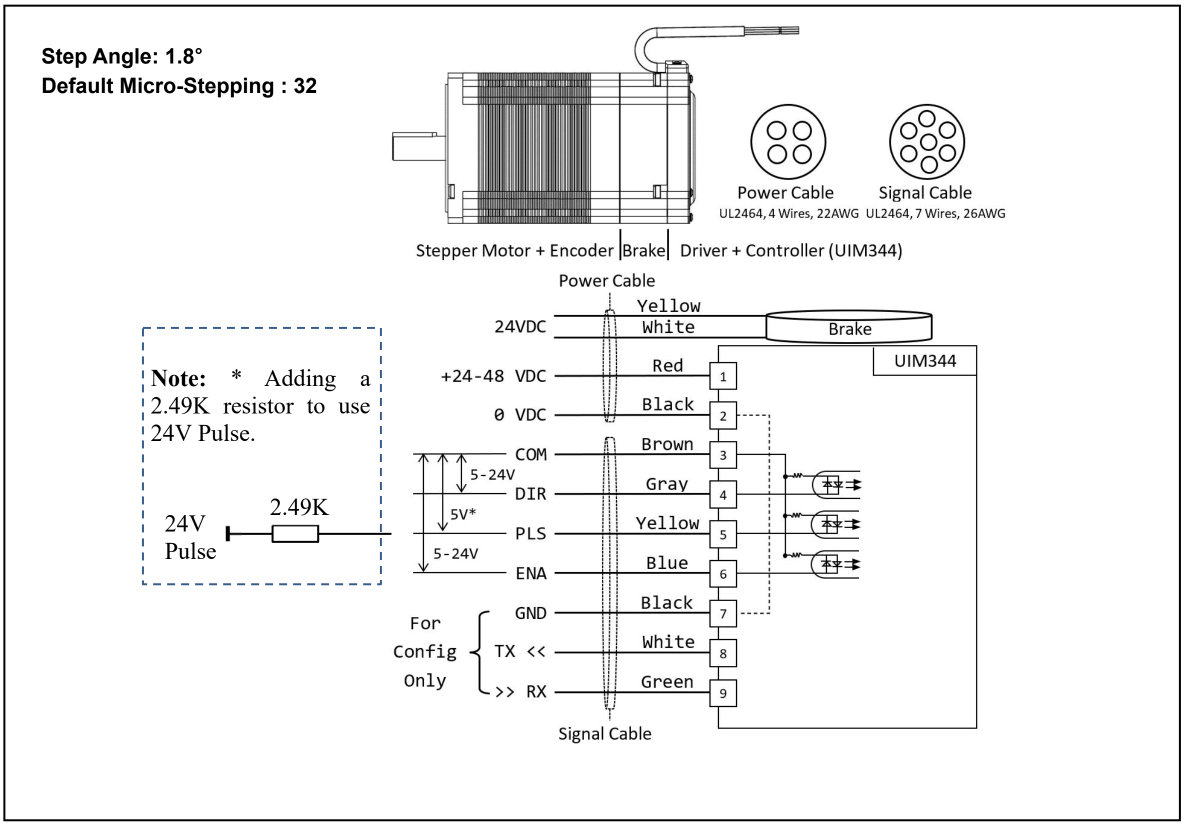


Figure 0-9: UIM344 Wiring Diagram with Brake (Motor Model: UIM8696PMB)

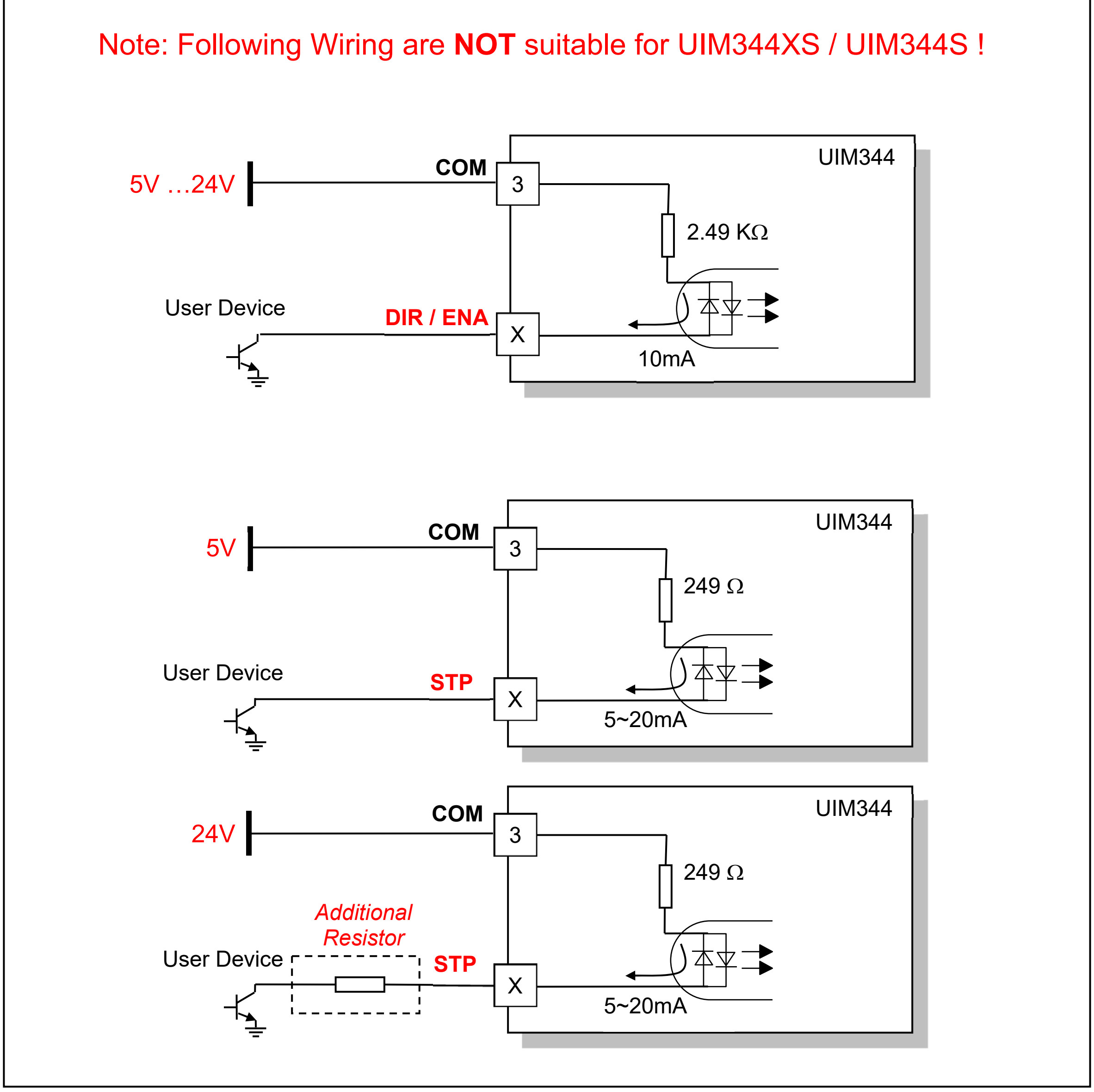


Optically Isolated Input Interface

UIM344 controllers' logic inputs are all optically isolated. All opto-isolators share one common anode (COM) as shown in above schematic diagram. Typically, COM is 5V. However, 3.3V or voltages higher than 5V are also acceptable, so long as the current through the opto-isolator's emitter is between 5~20mA.

Should a voltage higher than 5V (e.g., 24V) be applied to VCC, an additional resistor is needed for every terminal to ensure that the current through each emitter does not exceed 20mA.

Figure 0-10: Optically Isolated Input Interface



Characteristics

Absolute Maximum Ratings

Supply Voltage.. 10V to 50V
Ambient temperature under bias………….. -40°C to +85°C
Storage temperature... -50°C to +150°C

NOTICE: Stresses above those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device at those or any other conditions above those indicated in the operation listings of this specification is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Electrical Characteristics (Ambient Temperature 25°C)

Supply Power Voltage	12 – 48VDC
Motor Output Current	Max 2A/4A/8A per phase (Work/Idel Current can be set through UART)
Driving Mode	PWM constant current
Micro Stepping Resolution	1, 1/2, 1/4, 1/8, 1/16, **1/32 (Default)**, 1/64, 1/128 (Set through UART)
Control Interface	3-wire interface: Pulse, Direction, Shutdown
Insulation Resistance	>100MΩ
Dielectric Strength	0.5KV in one minute

Environmental Requirements

Cooling	Free air
Working environment	Avoid dust, oil mist and corrosive gases
Working temperature	-40°C ~ 85°C
Humidity	<80%RH，no condensation, no frosting
Vibration	3G Max
Storage temperature	-50°C ~ 150°C

Size and Weight

Controller Size	20 x 20 x 16 mm / 28 x 28 x 8 mm / 42 x 42 x 16.5 mm (L*W*H)
Wight	< 0.1 kg

LED Indicator Patterns and Meanings

Major LED indicator patterns are listed below:

Driver Parameter Configuration

Following parameters can be configured by user:

1) **Micro-Stepping resolution :** available values are 1, 2, 4, 8,16, 32, 64 and 128.
2) **Working current :** the current provided by the drive to the motor when running.
3) **Idle current rate :** the current applied when motor is not turning. Idle current is set in the form of percentage of the working current.
4) **Maximum Missing Steps :** Exceed this value, the motor will stop and locked.

These parameters can be read and set through UART as shown in the figure below.

Figure 0-11: Wiring Diagram for Configuration (UIM344XS)

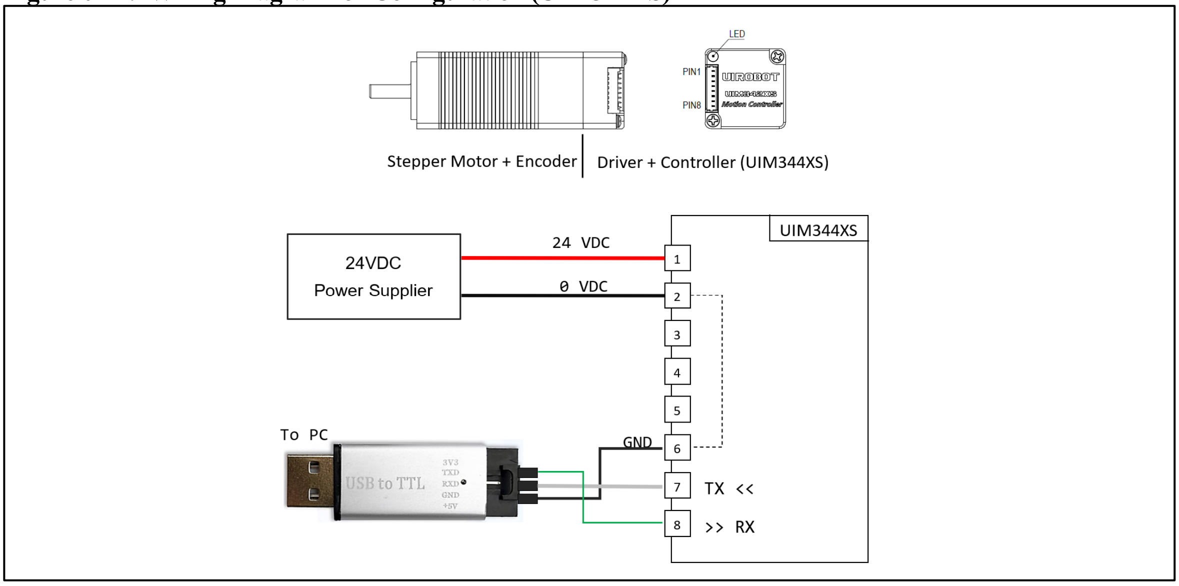


Figure 0-12: Wiring Diagram for Configuration (UIM344S)

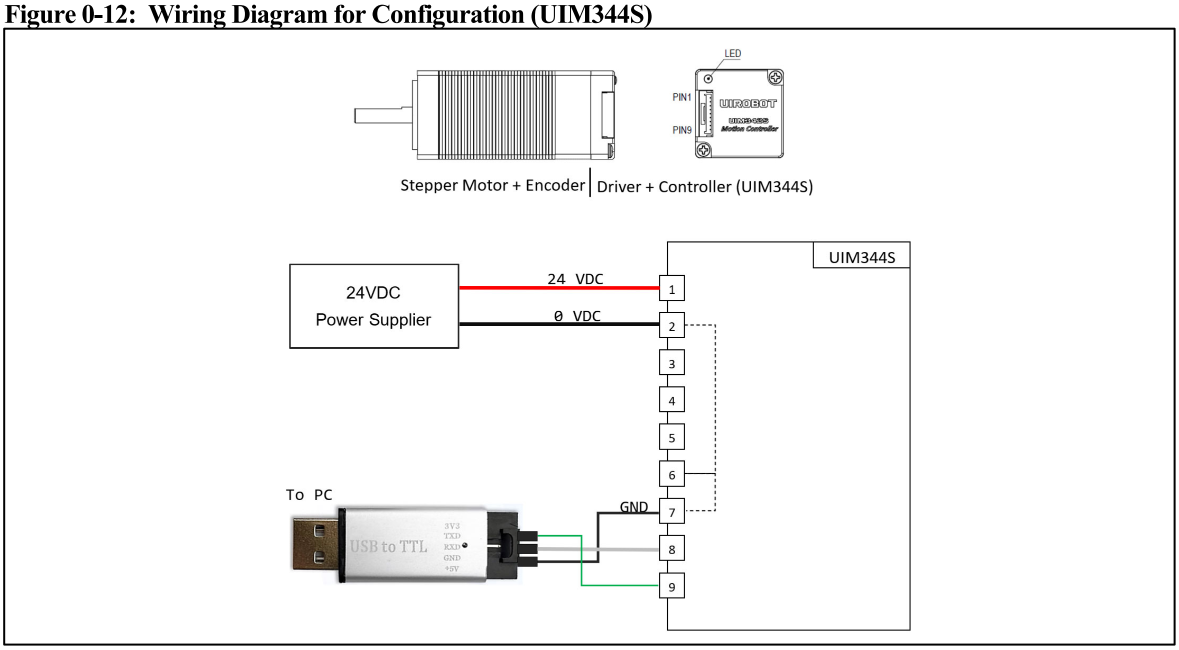


Figure 0-13: Wiring Diagram for Configuration (UIM4247PM)

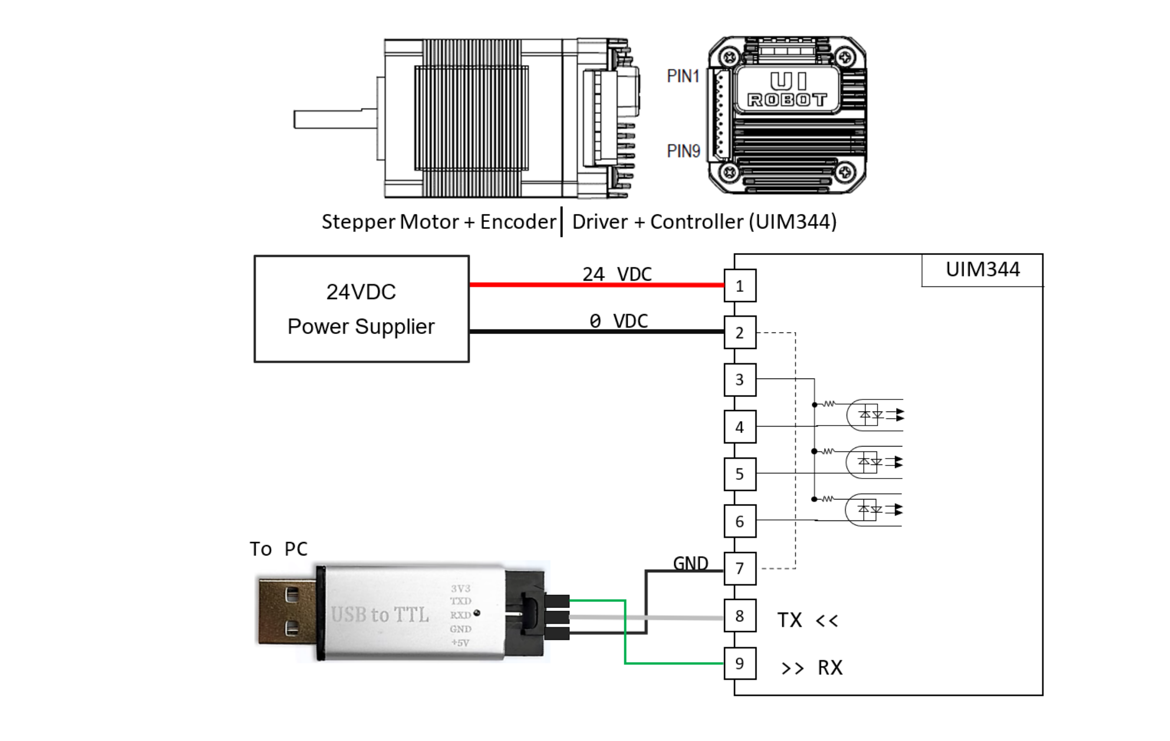
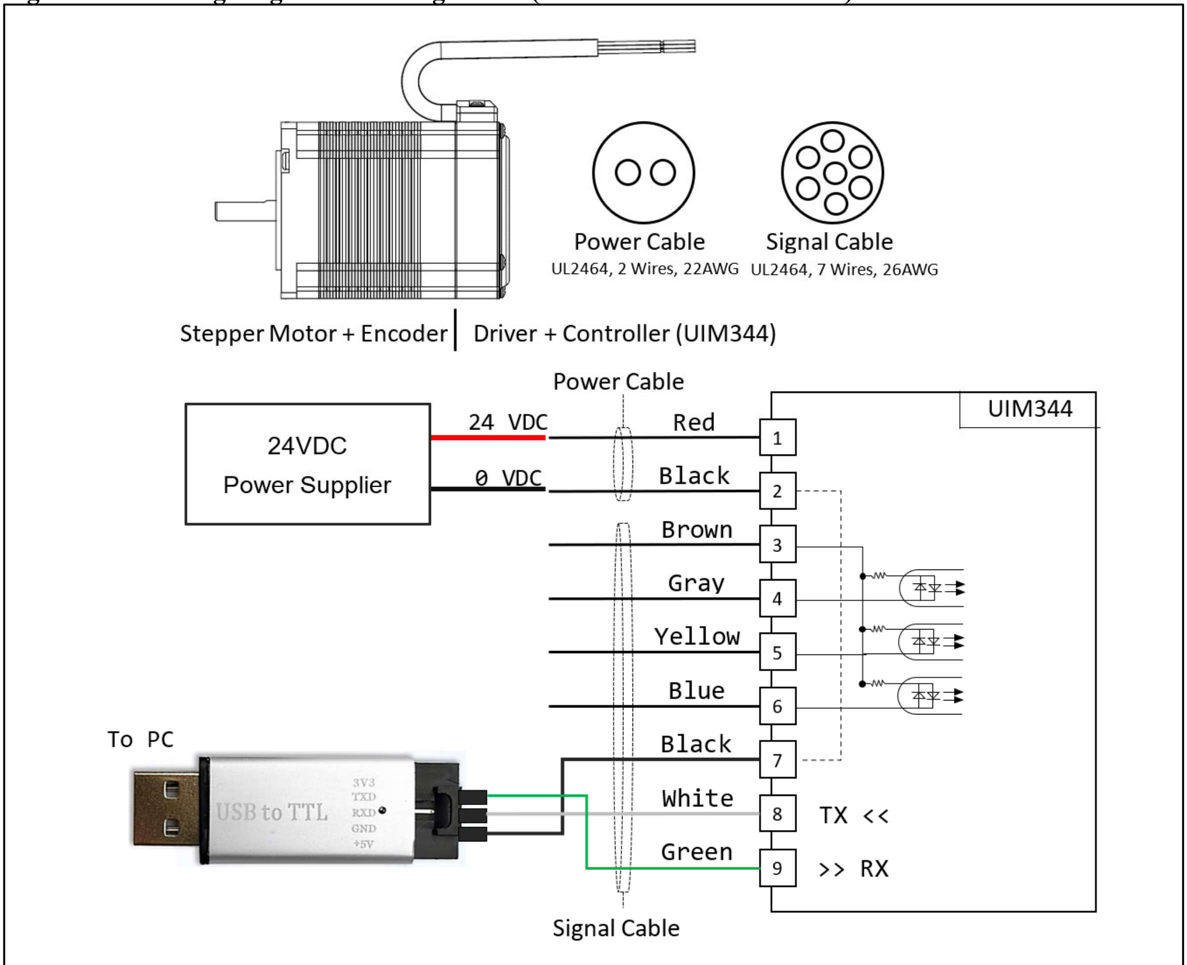


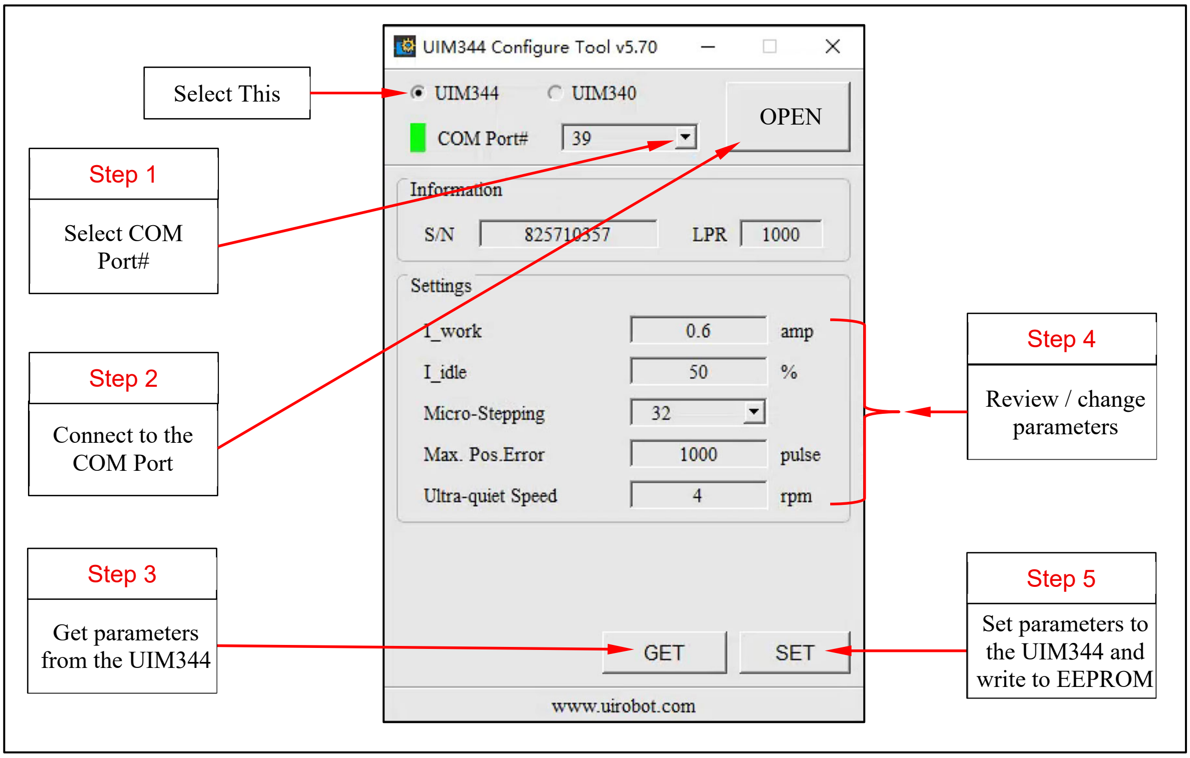
Figure 0-14: Wiring Diagram for Configuration (UIM5756PM / UIM8696PM)



After correctly connecting the PC (windows), USB-UART converter and the UIM344, please open the UIROBOT CFG344 (UIM344 Config Tool), which can be downloaded from UIROBOT's website www.uirobot.com.

Follow the steps shown in Figure 0-10 to configure.

Figure 0-15: UIM344 Config Tool



Note:

1) This is a UART interface, NOT RS232.
2) Do NOT change parameters when the motor driver is ON.
3) Motor needs to be powered.
4) The motor needs to reboot after configuration.